CSE 107 — Midterm 2 Prep

Practice paper

Fall 2026

65 minutes | 100 points

Solutions and rubric

Designed for 55 minutes of work; 65 minutes allowed.

Grading. Award the stated method and result points. Accept exact unsimplified numerical expressions; required integrals must be evaluated. Accept equivalent methods consistent with explicit requirements in the question. Carry arithmetic errors forward when subsequent work correctly uses the earlier result; deduct new errors separately. Accept complete labeled PMF tables or equivalent formulas with supports and zero elsewhere. PDF values at individual endpoints and inclusion of boundaries in sketches do not affect credit. CDF values must be correct at every real input, including endpoints.

1. (16 points)

A correct T earns 2 points without explanation. A correct F earns 1 point, plus 1 for a valid reason, correction, or counterexample. An incorrect selection earns 0. The key is **F, T, F, T, T, F, T, F**.

(a) **False.** Independence is not given. For example, let every I_j equal the same Bernoulli$(1/3)$ variable I. Then the sum is $6I$, which takes only 0 and 6, not a Binomial$(6, 1/3)$ law. **[2 points]**

Rubric. 1 for F; 1 for identifying missing independence or giving a valid counterexample.

(b) **True.** Adding constants does not change deviations from the mean. Their expected product, the covariance, is therefore unchanged. **[2 points]**

(c) **False.** Standard deviation scales by the absolute value of the coefficient. Hence $\mathrm{SD}(5 - 2X) = 2\,\mathrm{SD}(X)$. **[2 points]**

Rubric. 1 for F; 1 for the correct positive factor or the rule $\mathrm{SD}(aX + b) = |a|\,\mathrm{SD}(X)$.

(d) **True.** A continuous random variable assigns probability zero to each endpoint, so adding or removing endpoints does not change an interval probability. **[2 points]**

(e) **True.** Independence gives $p_{R|S}(r \mid s) = p_R(r)$ when $P(S = s) > 0$. The conditional and unconditional means therefore use the same weights. **[2 points]**

(f) **False.** A conditional density is normalized: $\int_{-\infty}^{\infty} f_{X|Y}(x \mid y)\,dx = 1$. The joint density integrates over x to $f_Y(y)$, so dividing by the positive $f_Y(y)$ gives total integral 1. **[2 points]**

Rubric. 1 for F; 1 for correcting the integral to 1 or explaining conditional normalization.

(g) **True.** The general variance formula has the extra term $2\,\mathrm{Cov}(X, Y)$. A zero covariance removes that term; independence is not needed. **[2 points]**

(h) **False.** The factorization holds for all r, s exactly when R and S are independent. For example, take $R = S \sim$ Bernoulli$(1/2)$. Then $p_{R,S}(1, 1) = 1/2$, but $p_R(1)p_S(1) = 1/4$. **[2 points]**

Rubric. 1 for F; 1 for stating the required independence condition or giving a valid counterexample.

2. (24 points)

(a) Joint and marginal PMFs. [6 points]

Both row probabilities are 1/2. Hence $p_{R,S}(r,s) = \frac{1}{2}p_{S|R}(s \mid r)$.

$p_{R,S}(r,s)$	$s=-1$	$s=0$	$s=1$	$p_R(r)$
$r=0$	1/8	1/4	1/8	1/2
$r=1$	1/4	0	1/4	1/2
$p_S(s)$	3/8	1/4	3/8	1

The joint PMF is zero outside $\{0,1\} \times \{-1,0,1\}$ and also at $(1,0)$. The marginal PMF of S is given in the last row and is zero elsewhere.

Rubric. Joint PMF, 4: 1/2 for each of the six cells (3 total), and 1 for the complete support and zero elsewhere. Marginal of S, 2: 1 for summing columns and 1 for the correct complete PMF.

(b) Reverse conditioning and independence. [6 points]

The event $S \geq 0$ contains the columns $s=0$ and $s=1$, so $P(S \geq 0) = 1/4 + 3/8 = 5/8$. Thus

$$P(R=r \mid S \geq 0) = \begin{cases} (1/4+1/8)/(5/8) = 3/5, & r=0, \\ (0+1/4)/(5/8) = 2/5, & r=1, \\ 0, & \text{otherwise.} \end{cases}$$

The variables are dependent, since

$$P(R=1 \mid S \geq 0) = \frac{2}{5} \neq \frac{1}{2} = P(R=1).$$

Alternatively, $p_{R,S}(1,0) = 0 \neq (1/2)(1/4)$. One decisive comparison is sufficient.

Rubric. Conditional PMF, 3: 1 for the denominator from both event columns, 1 for the probabilities, and 1 for complete support. Independence, 3: 1 for the decision and 2 for a valid comparison.

(c) Means and variances. [6 points]

For R, use the Bernoulli formulas. For S, use its three-point marginal:

$$E[R] = \frac{1}{2}, \qquad \mathrm{Var}(R) = \frac{1}{2}\left(1-\frac{1}{2}\right) = \frac{1}{4},$$
$$E[S] = -\frac{3}{8} + \frac{3}{8} = 0, \qquad E[S^2] = \frac{3}{8} + \frac{3}{8} = \frac{3}{4}.$$

Therefore $\mathrm{Var}(S) = E[S^2] - (E[S])^2 = 3/4$.

Rubric. For each variable: 1 for the mean, 1 for a correct variance formula, and 1 for its value; 3 points per variable.

(d) Product moment, correlation, and dependence. [6 points]

Only the row $r=1$ contributes to the product moment:

$$E[RS] = (-1)\cdot\frac{1}{4} + 0 + 1\cdot\frac{1}{4} = 0, \qquad \mathrm{Cov}(R,S) = 0 - \frac{1}{2}(0) = 0.$$

$$\mathrm{Corr}(R,S) = \frac{0}{\sqrt{1/4}\sqrt{3/4}} = 0.$$

Zero correlation does not imply independence. These variables are uncorrelated but dependent, as the comparison in (b) establishes; the two conclusions are consistent.

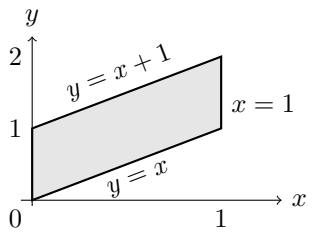
Rubric. 2 for $E[RS]$ (weighted sum and value); 1 for covariance; 1 for correlation with a valid nonzero denominator; 2 for explaining that zero correlation is consistent with dependence.

3. (28 points)

(a) Support. [4 points]

The positive-density region is the interior of the parallelogram with vertices $(0,0)$, $(0,1)$, $(1,2)$, and $(1,1)$. Its boundaries are $x = 0$, $x = 1$, $y = x$, and $y = x + 1$.

Rubric. 2 for the correct region; 2 for its boundary labels. Equivalent clear axis/vertex markings are accepted. No normalization calculation is required.



(b) Marginal density of Y. [8 points]

A horizontal slice must satisfy $0 < x < 1$ and $y - 1 < x < y$. The limits change at $y = 1$:

$$f_Y(y) = \begin{cases} \displaystyle\int_0^y 2x\,dx = y^2, & 0 < y < 1, \\ \displaystyle\int_{y-1}^1 2x\,dx = 1 - (y-1)^2 = 2y - y^2, & 1 \le y < 2, \\ 0, & \text{otherwise.} \end{cases}$$

Rubric. For each branch, 3: correct x-limits 1, integration 1, formula 1 (6 total). Complete y-domains and zero elsewhere 2. Equivalent evaluated formulas using minimum/maximum limits are accepted.

(c) Conditional density and mean. [6 points]

For each fixed $0 < x < 1$, divide by the supplied marginal $f_X(x) = 2x$:

$$f_{Y|X}(y \mid x) = \begin{cases} 1, & x < y < x + 1, \\ 0, & \text{otherwise,} \end{cases} \qquad E[Y \mid X = x] = \int_x^{x+1} y\,dy = x + \frac{1}{2}.$$

Rubric. Density, 3: ratio 1, value 1, y-support and zero elsewhere 1. Mean, 3: a valid integral or uniform-mean formula 1, correct result 2.

(d) Probability of $Y > 2X$. [6 points]

For $0 < x < 1$, the event boundary satisfies $x < 2x < x + 1$. Therefore

$$P(Y > 2X) = \int_0^1 \int_{2x}^{x+1} 2x\,dy\,dx = \int_0^1 2x(1-x)\,dx = 1 - \frac{2}{3} = \frac{1}{3}.$$

Rubric. 2 for valid event limits; 2 for the correct integrand; 2 for evaluation. Either integration order is accepted; no additional sketch is required.

(e) Independence. [4 points]

They are dependent. The conditional distribution is uniform on $(x, x+1)$, so its support changes with x. For example, $P(Y < 1 \mid X = x) = 1 - x$ for $0 < x < 1$; under independence this conditional probability would be constant in x.

Rubric. 1 for the conclusion; 3 for a valid conditional-law, probability, or joint/marginal-density comparison. The constant conditional-density value alone does not establish independence: its support must also be considered.

4. (16 points)

(a) Complete CDF by the CDF method. [8 points]

$0 < U < 1$ implies $0 < V < 1$. For $0 < v < 1$, the positive quantity $1 - U$ gives

$$\{(1-U)^2 \le v\} = \{1 - U \le \sqrt{v}\} = \{U \ge 1 - \sqrt{v}\}.$$

Integrating the density of U over this interval gives

$$F_V(v) = \int_{1-\sqrt{v}}^{1} 2u\,du = 1 - (1-\sqrt{v})^2 = 2\sqrt{v} - v.$$

The complete CDF is

$$F_V(v) = \begin{cases} 0, & v \le 0, \\ 2\sqrt{v} - v, & 0 < v < 1, \\ 1, & v \ge 1. \end{cases}$$

Rubric. Correct input event and inequality 3; probability calculation 2; interior CDF domain 1; outside branches and endpoint values 2. No separate range statement is required.

(b) Density. [2 points]

Differentiating the interior CDF gives

$$f_V(v) = \begin{cases} v^{-1/2} - 1, & 0 < v < 1, \\ 0, & \text{otherwise.} \end{cases}$$

The CDF is continuous at both endpoints. Although the density grows without bound near zero, it is integrable and $P(V = 0) = 0$.

Rubric. 1 for the derivative; 1 for the domain and zero elsewhere. The formula need not be evaluated at zero.

(c) Mean and variance. [6 points]

Using the density of V,

$$E[V] = \int_0^1 v\left(v^{-1/2} - 1\right)\,dv = \frac{2}{3} - \frac{1}{2} = \frac{1}{6}.$$

With the supplied second moment,

$$\mathrm{Var}(V) = E[V^2] - (E[V])^2 = \frac{1}{15} - \left(\frac{1}{6}\right)^2 = \frac{7}{180}.$$

Integrating with the density of U is also valid:

$$E[V] = \int_0^1 (1-u)^2\,2u\,du = 2\left(\frac{1}{2} - \frac{2}{3} + \frac{1}{4}\right) = \frac{1}{6}.$$

Either calculation earns full credit.

Rubric. 2 for a correct mean integral using either density; 1 for its value; 2 for the variance identity using 1/15 and the computed mean; 1 for a correct final variance. Exact numerical expressions need not be simplified. No second-moment integral is required.

5. (16 points)

(a) Normal probability. [3 points]

Let $Z_0 = (G - 30)/4 \sim N(0, 1)$, since the standard deviation is $\sqrt{16} = 4$. Then

$$P(|G - 30| \le 6) = P(-3/2 \le Z_0 \le 3/2) = \Phi(3/2) - \Phi(-3/2) = 2\Phi(3/2) - 1.$$

Rubric. 1 for standardizing by 4; 1 for the two bounds; 1 for a correct expression using Φ.

(b) Exponential conditional probability. [4 points]

Mean 3 minutes gives rate $\lambda = 1/3$ per minute and $P(T > t) = e^{-t/3}$ for $t \ge 0$. Because $\{T > 5\} \subset \{T > 2\}$,

$$P(T > 5 \mid T > 2) = \frac{P(\{T > 5\} \cap \{T > 2\})}{P(T > 2)} = \frac{e^{-5/3}}{e^{-2/3}} = e^{-1}.$$

Rubric. 1 for the rate; 2 for the conditional-probability ratio with the correct nested-event numerator; 1 for the result. No named memorylessness property or proof is required.

(c) Mean, variance, and SD of an average. [5 points]

The standard uniform formulas give

$$E[U_i] = \frac{-1 + 3}{2} = 1, \qquad \mathrm{Var}(U_i) = \frac{(3 - (-1))^2}{12} = \frac{4}{3}.$$

By linearity, $E[\overline{U}] = (1/8)\sum_{i=1}^{8} E[U_i] = 1$. Independence makes the cross-covariances zero, so

$$\mathrm{Var}(\overline{U}) = \frac{1}{8^2}\sum_{i=1}^{8} \mathrm{Var}(U_i) = \frac{8(4/3)}{64} = \frac{1}{6}, \qquad \mathrm{SD}(\overline{U}) = \frac{1}{\sqrt{6}}.$$

Rubric. 1 for the average's mean; 1 for the individual uniform variance; 2 for the average's variance, including where independence is used; 1 for its SD. Uniform moment formulas may be used without derivation.

(d) A linear combination with dependence. [4 points]

Convert correlation to covariance:

$$\mathrm{Cov}(W, Z) = \mathrm{Corr}(W, Z)\sqrt{\mathrm{Var}(W)}\sqrt{\mathrm{Var}(Z)} = -\frac{1}{4}(1)(2) = -\frac{1}{2}.$$

The constant shift contributes no variance. Keeping the cross term gives

$$\begin{aligned}\mathrm{Var}(3W + 2Z - 7) &= 3^2\,\mathrm{Var}(W) + 2^2\,\mathrm{Var}(Z) + 2(3)(2)\,\mathrm{Cov}(W, Z) \\ &= 9 + 16 + 12\left(-\frac{1}{2}\right) = 19.\end{aligned}$$

Rubric. 1 for converting correlation to covariance; 1 for the squared-coefficient variance terms; 1 for the covariance term; 1 for the final variance.